Stability-Penalized Split Selection for Decision Tree Classification: A Reproducible Multi-Dataset Evaluation

Rofiq Fauzi
Magister Artificial Intelligence
Universitas Dian Nuswantoro
Semarang, Indonesia
p36202600047@mhs.dinus.ac.id

***Abstract*—The CART decision tree is fast and interpretable, but its structure is unstable: small changes in the training data can alter the selected split and, ultimately, the predictions. This study evaluates a simple candidate formulation, named Stability-Penalized Decision Tree (SPDT), in which each candidate split is scored as Gini gain minus a weighted penalty on the variance of that gain over bootstrap resamples of the node data. A literature search up to 22 October 2026 found no formula-level equivalent of this criterion, but the tree-stability research space is crowded; the formulation is therefore positioned as a candidate, not as a novelty claim. The evaluation uses eight UCI tabular classification datasets with repeated nested cross-validation (5 folds $\times$ 3 repeats on three seeds), hyperparameter tuning in the inner loop, and Macro-F1 as the primary metric. Results: SPDT versus custom CART (λ=0) records six wins, zero ties, and two losses with a mean Macro-F1 difference of $+0.0058$. The Friedman test gives $\chi^2(2)$=4.0000, p=0.1353, Kendall's W=0.2500 — insufficient evidence to reject the null hypothesis, so confirmatory post-hoc comparisons were not run. Ablations show that the penalty strength λ and the number of resamples B barely affect Macro-F1, tree depth constraints matter far more, and the prediction-stability measure of SPDT is slightly worse than that of custom CART at roughly $40\times$ the fitting time. These findings are an honest null result: across the eight small-to-medium datasets tested, the bootstrap gain-variance penalty provides no benefit sufficient to outweigh the simplicity of CART, without ruling out benefits in other data regimes.**

***Index Terms*—Decision trees, split criterion, bootstrap stability, gain variance, nested cross-validation, nonparametric statistical comparison.**

I. Introduction

DECISION trees are nonparametric classifiers that recursively partition the feature space. The CART split criterion selects the candidate (feature, threshold) that maximizes the reduction of Gini impurity [1], [2]. However, tree structure is known to be unstable: small perturbations of the training data can change the selected split [3], and consequently the predictions — the very phenomenon behind bagging and random forests [4]–[6].

Various approaches have reduced this instability at different levels: significance filtering, structural restriction, ensemble aggregation, or stable model selection [7]–[10]. What is rarely tested directly is the narrower question: **can a bootstrap variance penalty on the split gain, subtracted directly from the split score, improve out-of-sample Macro-F1 and/or prediction stability compared with standard CART under repeated resampling, at acceptable runtime cost?**

Null hypothesis: no systematic improvement over the baseline across independent datasets. No hypothesis is made that SPDT must win. The claimed contributions are limited to (i) an explicit candidate formulation, (ii) a reproducible implementation and audit, (iii) a leakage-free nested-CV evaluation, and (iv) honest ablation and statistical analysis of both positive and negative outcomes.

II. Related Work

A. Decision Tree Stability

Structural tree instability was formalized by Li and Clarke [3]. Quantitative stability and tree-similarity measures were developed by Turney [7] and Briand et al. [8], who also evaluated bootstrap-based node-level stabilization (the NLS/Dannegger lineage). Lee et al. [9] proposed ANLS, which runs bootstrap iterations at each node and selects the most frequently chosen predictor — the closest resampling-per-node mechanism to this study, but its aggregation is frequency voting rather than a gain-variance penalty. Mirzamomen and Kangavari [10] stabilize trees through multi-attribute split tests.

B. Alternative Split Criteria

Two-term Tsallis entropy criteria [11], label-noise-robust criteria [12], and bias-corrected Gini split selection [13] show that the "modify the split score" space is active. Hoeffding-bound data-stream trees [14] decide splits only when the gain estimate is separated from the runner-up by a confidence interval, and gradient-boosted trees regularize splitting with a minimum-split-gain threshold [15]. Delise [16] scores split consistency across environments in boosting. Penalizing a resampled score by its variance is also known outside trees (the mean–variance pattern in model selection). None of the works we verified contains the formula $\mathrm{Gain} - \lambda\,\mathrm{Var}_{\mathrm{bootstrap}}[\mathrm{Gain}]$; nevertheless, a negative search result is not proof of novelty, and the formulation remains a candidate.

III. Method

A. Proposed Split Criterion

For a node with dataset D (training portion only), class probabilities $p(c \mid D)$, and C classes, the Gini impurity is

$$\mathrm{Gini}(D) = 1 - \sum_{c=1}^{C} p(c \mid D)^2. \tag{1}$$

For a candidate split s partitioning D into D_L and D_R, the CART gain is

$$\mathrm{Gain}(s, D) = \mathrm{Gini}(D) - \tfrac{|D_L|}{|D|}\,\mathrm{Gini}(D_L) - \tfrac{|D_R|}{|D|}\,\mathrm{Gini}(D_R). \tag{2}$$

From D we draw B bootstrap resamples $D^{(b)}$ (sampling with replacement, node rows only). With the candidate (s) fixed, the resampled gain variance is

$$V(s) = \mathrm{Var}_b\Big[\mathrm{Gain}(s, D^{(b)})\Big], \qquad b = 1, \ldots, B, \tag{3}$$

and the proposed score is

$$\boxed{\mathrm{Score}(s) = \mathrm{Gain}(s, D) - \lambda\, V(s)} \tag{4}$$

where $\lambda \geq 0$ controls the penalty strength. Here D is the current node's training data, s is a (feature, threshold) candidate, B is the number of bootstrap resamples, and λ weights the penalty; λ=0 disables the penalty entirely. Candidate thresholds are midpoints between sorted distinct adjacent values; $\mathrm{min_samples_leaf}$ and $\mathrm{min_samples_split}$ are applied to original node counts; an empty child partition inside a resample contributes zero; leaves predict the majority class with smallest-label tie-breaking; bootstrap seeds derive deterministically from $(\mathrm{random_state}, \mathrm{node\ id})$. For λ=0 the estimator degenerates to a custom CART reference, verified identical to an independent reference implementation in unit tests. A separate exploratory variant normalizing $V(s)$ by $\mathrm{Gain}(s, D)^2 + \varepsilon$ is documented but is not the main method. An important limitation is that $V(s)$ is scale-dependent on the gain.

B. Evaluation Design

We use eight UCI tabular classification datasets (Table I). The protocol is repeated stratified nested CV: an outer loop of 5 folds $\times$ 3 repeats for each of the seeds $[42, 123, 2026]$ (45 evaluations per dataset–method, identical splits for all methods), and an inner stratified 3-fold loop for tuning with deterministically derived seeds. Preprocessing is median imputation fitted on the training portion only. The primary metric is Macro-F1; secondary measures are balanced accuracy, accuracy, fit/predict time, node count, and tree depth. Prediction stability is measured separately as mean pairwise disagreement among models trained on ten 90% subsamples of each outer training fold, evaluated on the same outer test fold, and is reported separately from accuracy. The primary comparison set is sklearn CART, custom CART (λ=0), and SPDT; a random forest serves as a non-equivalent contextual benchmark outside the omnibus test.

TABLE I
Dataset Inventory

Dataset	N	Features	Classes	Source
banknote	1372	4	2	ucimlrepo
glass	214	9	6	ucimlrepo
ionosphere	351	34	2	ucimlrepo
iris	150	4	3	ucimlrepo
seeds	210	7	3	UCI zip
sonar	208	60	2	ucimlrepo
wdbc	569	30	2	ucimlrepo
wine	178	13	3	ucimlrepo

C. Statistical Analysis

The unit of inference is the dataset (averaged over the 45 outer folds), never the individual fold. The three primary methods are compared with the Friedman test using datasets as blocks; only when the omnibus test provides grounds for it are paired Wilcoxon signed-rank tests with Holm correction and rank-biserial effect sizes run. $p \geq 0.05$ is never interpreted as evidence of equivalence.

IV. Results and Discussion

Table II reports per-dataset Macro-F1 (mean $\pm$ SD over the 45 outer folds, descriptive only). Table III reports mean ranks, and Table IV reports the omnibus test. SPDT versus custom CART records six wins, zero ties, and two losses with mean Macro-F1 difference $+0.0058$. The Friedman test gives $\chi^2(2) = 4.0000$, $p = 0.1353$, Kendall's $W = 0.2500$ — insufficient evidence to reject the null hypothesis, so confirmatory post-hoc comparisons were not run. The exploratory SPDT–custom CART Wilcoxon comparison gives a raw $p = 0.078$, which does not survive the 5% level and must not be read as significant; with only eight datasets, statistical power is limited. These results are reported as they stand.

A second finding concerns stability. The bootstrap-variance penalty does *not* reduce prediction disagreement: SPDT is slightly worse (0.101) than custom CART (0.093) and sklearn CART (0.098), while the random forest remains the most stable (0.027) through ensemble aggregation (Table VI). The motivating promise of the method — better prediction stability — therefore does not materialize on this benchmark; a gain-variance penalty is no substitute for ensembling.

Third, the identical-split ablations (Table VII) show that neither the penalty strength λ nor the number of resamples B materially changes Macro-F1 (all λ and B variants fall in the 0.859–0.861 range), whereas depth/leaf constraints matter far more (depth 3 drops to about 0.836). Inner tuning frequently selects a large λ (10 on 64.7% of folds) and a small B (5 on 53.1%), per Table VIII, indicating the penalty is often selected but has a small effect on accuracy. The cost is real: SPDT fitting time is roughly $40\times$ that of custom CART (Table V). Overall, the effectiveness of the bootstrap gain-variance penalty is dataset-dependent and small in magnitude; simple impurity-based splitting remains preferable for these small-to-medium datasets.

TABLE II
MACRO-F1 MEAN ± SD (DESCRIPTIVE FOLD-LEVEL SCORES) PER DATASET AND METHOD

Dataset	CART-skl	CART-cust	SPDT	RF	SPDT−CART
banknote	0.982 ± 0.008	0.981 ± 0.007	0.981 ± 0.007	0.994 ± 0.004	+0.0004
glass	0.620 ± 0.097	0.609 ± 0.093	0.626 ± 0.093	0.742 ± 0.075	+0.0166
ionosphere	0.873 ± 0.034	0.881 ± 0.038	0.879 ± 0.038	0.929 ± 0.037	-0.0027
iris	0.949 ± 0.033	0.945 ± 0.038	0.944 ± 0.037	0.955 ± 0.034	-0.0008
seeds	0.897 ± 0.032	0.906 ± 0.036	0.913 ± 0.034	0.923 ± 0.038	+0.0066
sonar	0.719 ± 0.065	0.706 ± 0.073	0.725 ± 0.061	0.833 ± 0.065	+0.0189
wdbc	0.925 ± 0.031	0.925 ± 0.030	0.928 ± 0.030	0.956 ± 0.023	+0.0037
wine	0.905 ± 0.047	0.905 ± 0.044	0.909 ± 0.046	0.981 ± 0.024	+0.0040

TABLE III
MEAN MACRO-F1 RANK ACROSS DATASETS

Method	Mean rank
SPDT	1.500
CART-skl	2.000
CART-cust	2.500

TABLE IV
DATASET-LEVEL STATISTICAL TEST (FRIEDMAN, D=8 DATASETS, 3 METHODS)

Statistic	Value
χ^2	4.0000
Degrees of freedom	2
p	0.1353
Kendall's W	0.2500
Decision	post-hoc not run

TABLE V
COMPUTATIONAL COST (PER OUTER FOLD, AVERAGED OVER DATASETS)

Method	Fit (s)	Fit median (s)	Predict (s)	Nodes	Depth
CART-cust	0.5340	0.4163	0.0009	26.6722	5.1000
CART-skl	0.2693	0.2605	0.0010	27.4222	5.2917
RF	1.1169	1.0863	0.0573	-	-
SPDT	21.4302	14.6491	0.0010	28.5944	5.4611

TABLE VI
PREDICTION STABILITY (MEAN PAIRWISE DISAGREEMENT; NOT AN ACCURACY METRIC)

Method	Mean disagreement	SD
CART-cust	0.0928	0.0853
CART-skl	0.0983	0.0885
RF	0.0271	0.0337
SPDT	0.1014	0.0934

TABLE VII
ABLATION ON IDENTICAL OUTER SPLITS

Variant	Macro F1	SD	Fit (s)	Nodes
A0 (λ=0)	0.8599	0.1270	0.0201	37.6611
A1 (λ=.1)	0.8598	0.1256	0.0860	37.6111
A1 (λ=1)	0.8601	0.1255	0.0862	37.5278
A2 (B=20)	0.8601	0.1251	0.1136	37.4556
A2 (B=5)	0.8613	0.1243	0.0711	37.4667
A3 d3 l1	0.8357	0.1592	0.0371	12.8722
A3 d3 l5	0.8353	0.1546	0.0373	11.8611
A3 d8 l1	0.8592	0.1272	0.0821	35.2500
A3 d8 l5	0.8521	0.1381	0.0820	24.7389

TABLE VIII
SPDT HYPERPARAMETER SELECTION FREQUENCY (45 OUTER FOLDS PER DATASET)

Parameter	Value	Frequency	Share
max_depth	5.0	140	0.3889
max_depth	8.0	102	0.2833
max_depth	3.0	92	0.2556
min_samples_leaf	1.0	201	0.5583
min_samples_leaf	2.0	97	0.2694
min_samples_leaf	5.0	62	0.1722
min_samples_split	2.0	252	0.7000
min_samples_split	5.0	69	0.1917
min_samples_split	10.0	39	0.1083
n_bootstrap	5.0	191	0.5306
n_bootstrap	20.0	87	0.2417
n_bootstrap	10.0	82	0.2278
penalty_lambda	10.0	233	0.6472
penalty_lambda	0.0	61	0.1694
penalty_lambda	0.01	41	0.1139
penalty_lambda	1.0	22	0.0611
penalty_lambda	0.1	3	0.0083

V. CONCLUSION

This study evaluated the SPDT candidate formulation reproducibly, with leakage-free repeated nested CV, controlled ablation on identical splits, and dataset-level statistical analysis on eight UCI datasets. Descriptively, SPDT achieved the best mean rank and won six of eight paired comparisons against custom CART, but the Friedman test provided insufficient evidence of a difference ($\chi^2(2) = 4.0000$, $p = 0.1353$), so there is no confirmatory basis for claiming improvement. Moreover, the prediction-stability measure did not improve and was slightly worse, while compute cost rose roughly $40\times$. Under the tested conditions, the simplicity of CART remains preferable. Limitations: eight small datasets, Gini impurity only, a single disagreement-based stability proxy, a formula-level novelty claim that is not established, and the absence of the closest literature criteria (ANLS, minimax split) as comparators. The most valuable future work is direct comparison with node-level bootstrap stabilization from the literature, a theoretically grounded penalty normalization by gain scale, and evaluation on larger benchmarks.

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